Online Resource 2 to: *Shift-aware probabilities for random convolutional kernel time series classifiers via leverage-moderated leave-one-out calibration*

Experimental design (fixed before the full benchmark was run)

Committed to the repository before any non-development dataset was evaluated with the final protocol. Development exposure is disclosed in §7.

1. Hypothesis

For random-convolutional-kernel time-series classifiers (MiniRocket and relatives) with a ridge readout, standardising the ridge logits by the square root of the test point's leverage, $z(x)/\sqrt{h(x)}$, and fitting a single temperature and the ridge penalty on the *exact* leave-one-out pairs (z_{-i}, h_{-i}), yields

- H1 in-distribution probabilities at least as well calibrated (NLL, ECE, Brier) as the best existing closed-form readout (PreVal) and as cross-validated or hold-out calibration, with unchanged accuracy;
- H2 substantially better calibration and shift detection under test-time corruptions, because leverage grows for inputs off the training distribution;
- H3 closed-form jackknife+ prediction sets with near-nominal coverage on clean data and smaller sets than split conformal.

2. Research questions → experiments

RQ	Question	Experiment
RQ1	In-distribution calibration vs. baselines?	E1 main (clean), E2 strong baselines
RQ2	Which components matter?	E3 ablation
RQ3	Behaviour under test-time distribution shift?	E4 robustness (18 corruption conditions), shift-detection AUROC
RQ4	Does the gain transfer to other random-kernel extractors and dataset groups?	E5 generalisation (Rocket, MultiRocket, Hydra; dataset types; dev vs. non-dev)
RQ5	Cost?	E6 efficiency (fit/predict time, complexity)
RQ6	Sensitivity to design choices?	E7 (moderation strength c, smoothing, λ criterion, miscoverage α, training-set size)
RQ7	When does it fail?	E8 error analysis
RQ8	Valid prediction sets?	E9 conformal

3. Data

114 UCR univariate datasets with equal-length series and no missing values in the official text distribution (UCR 2018 archive; list in `code/data/raw/ucr_equal_length.csv`). Resamples r = 0 (official split) and r = 1…4 (stratified resamples of the pooled data that keep the official per-class training counts; numpy seed = r). The MiniRocket seed for resample r is 1000·r (+j for ensemble member j). No seed was chosen after seeing results.

4. Methods (all share identical MiniRocket features per resample)

Proposed: **LOOM** (standardised logits, Platt target smoothing, λ and a by LOO NLL). Baselines: Ridge-onehot (default output of ROCKET-family implementations), Ridge-softmax (T = 1), PreVal (Dempster et al.; global LOO scaling, faithful: no smoothing), Holdout-TS (80/20 temperature scaling), Ridge-Platt-CV5 and Ridge-Isotonic-CV5 (sklearn CalibratedClassifierCV), LogReg-CV (L2 logistic regression, C ∈ {1e-3,1e-2,1e-1,1} by 3-fold CV log-loss), PreVal-Ens5 and LOOM-Ens5 (average of five MiniRocket seeds), Arsenal and QUANT (aeon; raw-series classifiers with native probabilities).

Compute-driven protocol decisions (made after timing pilots, before any benchmark result): * LogReg-CV is skipped for datasets with more than 2,000 training series (13 fits of an L2 logistic regression with ~10,000 features per job); its comparisons use the remaining datasets. * Arsenal uses aeon's MiniRocket-kernel configuration (`rocket_transform="minirocket"`, 25 estimators x 2,000 kernels) instead of the default ROCKET kernels: the default configuration needed 32 s to predict 1,029 series of length 24, which extrapolates to several CPU-days for 19 test conditions on the archive. Arsenal is run on the official split (resample 0) only and is compared with the other methods on that split in a separate table. QUANT is run on all five resamples. Ablations: PreVal-smooth, LOOM-nosmooth, LOOM-free-b, LOOM-finite-c1, LOOM-kNN, LOOM-insample, LOOM-holdout, LOOM-lambdaSSE.

5. Corruptions (test set only; severities 1–3)

noise σ ∈ {0.1, 0.3, 0.5}·sd; wander amplitude ∈ {0.25, 0.5, 1}·sd; missing segment ∈ {10, 20, 30} % (linear interpolation); spikes ∈ {1, 3, 5} % of points at ±3 sd; smooth monotone time warp with slope ∈ [1−s, 1+s], s ∈ {0.05, 0.1, 0.2}; circular shift ∈ {5, 10, 20} % of the length. Seeds are a hash of (dataset, resample, family, severity): identical inputs for all methods.

6. Endpoints and statistics

- Primary endpoint: test NLL. Secondary: ECE (15 equal-width bins, top label), Brier, classwise ECE, accuracy, balanced accuracy, AURC, shift-detection AUROC, conformal coverage and set size (α = 0.1; 0.05 and 0.2 as sensitivity).
- Robustness endpoint: per-dataset mean over the 18 corruption conditions.
- Unit of analysis = dataset (metric averaged over resamples). Two-sided Wilcoxon signed-rank test LOOM vs. each competitor, Holm correction within each table, matched-pairs rank-biserial correlation, percentile-bootstrap 95 % CI (10,000 draws) for the median paired difference; Friedman test and average ranks over all methods.
- Significance level 0.05 after Holm correction. No claim of superiority without it.

Conformal scores: both LAC (1 − p) and deterministic APS (Romano et al. 2020) are reported for jackknife+ (LOOM, PreVal) and split conformal (Holdout-TS). APS was added after the quick-mode run on the six development-exposed datasets showed that LAC sets become empty when moderated probabilities are flat under shift; it was added before the full benchmark.

7. Development exposure (disclosed)

Method design used 8 randomly chosen development datasets (Adiac, Beef, Car, EthanolLevel, Fish, Lightning2, NonInvasiveFetalECGThorax2, SonyAIBORobotSurface1) with the official split and three ad-hoc corruptions. GunPoint, ECG200 and Coffee were used for pipeline smoke tests, and ItalyPowerDemand, SonyAIBORobotSurface2 and TwoLeadECG form the `run_all.py --mode quick` set; the GunPoint

smoke test revealed that separable LOO predictions drive the temperature to its bound, which motivated Platt target smoothing (decided before the benchmark). All results are reported for the 114 datasets and, as a sensitivity analysis, for the 100 datasets never seen during development, and for the 98 datasets without train/test duplicates (leakage audit, `code/results/leakage_audit.csv`).

8. Protocol amendments for compute (made before any benchmark result was inspected)

After the first full launch, one resample of the largest dataset (StarLightCurves) took 68 CPU-minutes with five-seed ensembles, and the container restarted, which terminated the run. Before any result file was opened, the following amendments were made and the four completed jobs were discarded: 1. PreVal-Ens5 and LOOM-Ens5 are computed on the official split (resample 0) only; resamples 1–4 use a single MiniRocket seed. Ensembles are compared with the other methods on the official split, in the same table as Arsenal and QUANT. 2. Conformal prediction sets are evaluated on a seeded random subset of at most 1,000 test series per dataset (the same subset for every set method and condition). 3. Rocket, MultiRocket and Hydra (generalisation) and Arsenal are evaluated on the official split, on clean data and the six severity-3 corruptions. 4. (Second amendment, also before inspecting results; 26 main jobs had finished.) Because the cloud container restarts every one to two hours and does not compute while idle, the Rocket extractor was dropped from the generalisation experiment (its transform costs roughly ten times MiniRocket's), keeping MultiRocket and Hydra, and the training-size experiment uses clean data and the six severity-3 corruptions. 5. (Third amendment, before inspecting results for these datasets.) For the two datasets with more than 3,000 training and more than 2,000 test series (Crop, ElectricDevices) the leverage projection costs about 10^13 operations per readout and seed. Their test sets are therefore evaluated on a seeded random subsample of 2,000 test series (identical for all methods and conditions). The other 112 datasets use their full test sets. 6. (Fourth amendment, before inspecting results of the affected datasets.) The worker running the largest datasets was killed twice by the out-of-memory killer (9 GB peak) and the remaining serial work on them was estimated at ~16 h. For the four datasets with more than 3,000 training series (Crop, ElectricDevices, FordA, FordB): the main experiment and QUANT use resamples 0–2 (Crop: 0–4, already completed); ElectricDevices is evaluated without the five-seed ensembles (single seed) and is therefore absent from the ensemble comparison; the extractor and training-size experiments exclude these four datasets. ElectricDevices and Crop jobs run alone at the end.

Note added 2026-10-21: the official-split ElectricDevices job had already finished with the five-seed ensembles before this amendment took effect, so ElectricDevices **is** included in the ensemble comparison (114 datasets); the single-seed rule affects only its resamples 1–2, which carry no ensembles for any dataset.

9. Post-hoc analysis changes (made after results were inspected; reported as such)

7. (Review round 8.) The Arsenal was run on clean and severity-3 data only (amendment 3), but the corrupted-data comparison of the official-split table originally averaged LOOM, PreVal, the ensembles and QUANT over all 18 conditions. This mixed different condition sets and favoured LOOM against the Arsenal. The official-split comparison (manuscript Table 3) now uses the six severity-3 conditions **for every method**. As a result QUANT becomes significantly better than LOOM in NLL on this table, which is reported in the abstract, results, discussion and conclusion. The Brier score was added to this table because the Arsenal's vote-based probabilities contain exact zeros that dominate its NLL. The primary endpoint and the main analysis (Tables 1–2, all 18 conditions, five resamples) are unchanged.
8. (Review round 7.) Fit times of Ridge-onehot/softmax now include the shared SVD they use, and the MiniRocket transform time is reported separately (it was recorded but not loaded because of a key mismatch). No benchmark job was rerun.

10. Application case study (designed after the main benchmark; protocol fixed before it was run, 2026-10-29)

Purpose. Evaluate the methods as they would be used in a monitoring application: a classifier labels incoming recordings automatically and passes the recordings it is least sure about to a human reviewer, while a fraction of recordings is affected by sensor faults. This study is **exploratory**: it was added after the main results were known, in response to the question of practical value, and it is reported as such.

Datasets (12, three application domains, chosen by domain before running; all UCR, official split plus resamples 1–4): cardiac monitoring (ECG200, ECG5000, TwoLeadECG, NonInvasiveFetalECGThorax1); wearable and motion sensing (UWaveGestureLibraryX, CricketX, GunPoint, SemgHandGenderCh2); industrial and energy sensing (Wafer, FreezerRegularTrain, PowerCons, RefrigerationDevices).

Fault model. Each test recording is independently faulty with probability 0.3. A faulty recording receives one fault drawn uniformly from the domain's plausible faults, at a severity drawn uniformly from {1, 2, 3}, using the corruption functions of the main benchmark: cardiac — broadband noise, baseline wander, electrode spikes, lead-off segment (missing); wearable — noise, spikes, packet loss (missing), speed variation (warp); industrial — noise, drift (wander), spikes, dropout (missing). All draws are seeded by (dataset, resample).

Methods. LOOM, PreVal, Holdout-TS, Ridge-Platt-CV5 (MiniRocket features, as in the benchmark) and QUANT.

Operating metrics (review budget $b \in \{10\ \%, 20\ \%, 30\ \%\}$; the b least confident recordings, by the maximum predicted probability, go to review): (1) error rate of the automatically accepted decisions; (2) share of faulty recordings caught by the review queue; (3) for reference, the same quantities on a fault-free stream. Primary operating point: b = 20 %. Secondary: NLL on the faulty stream.

Analysis. Dataset is the unit (mean over the five resamples); LOOM vs each method by two-sided Wilcoxon signed-rank tests with Holm correction over the four comparisons per metric; with 12 datasets these tests have low power, so per-domain medians are reported descriptively. Per-sample outputs are saved for re-analysis.

Table S1. Leakage audit of the official UCR splits

Exact duplicate series within and across the official splits for the 114 datasets; only datasets with at least one duplicate are listed (22 datasets; 16 with test series that duplicate a training series). The full table is `code/results/leakage_audit.csv`.

dataset	n_train	n_test	dup_within_train	dup_within_test	test_in_train	test_in_train_label_conflict
ArrowHead	36	175	0	1	0	0
Crop	7200	16800	680	2703	2753	0
ElectricDevices	8926	7711	169	251	208	101
FaceAll	560	1690	0	2	1	0
FacesUCR	200	2050	0	3	0	0

dataset	n_train	n_test	dup_within_train	dup_within_test	test_in_train	test_in_train_label_conflict
FreezerRegularTrain	150	2850	0	4	1	0
FreezerSmallTrain	28	2850	0	4	27	0
InlineSkate	100	550	1	29	6	0
InsectEPGSmallTrain	17	249	0	0	15	0
MedicalImages	381	760	0	0	1	0
MixedShapesRegularTrain	500	2425	0	5	1	0
MixedShapesSmallTrain	100	2425	0	5	0	0
PowerCons	180	180	35	35	67	0
Rock	20	50	0	1	0	0
ShapesAll	600	600	0	0	2	0
Strawberry	613	370	0	0	1	1
SwedishLeaf	500	625	0	1	2	0
Symbols	25	995	0	20	0	0
ToeSegmentation1	40	228	0	12	4	0
UMD	36	144	1	0	0	0
Worms	181	77	23	0	10	0
WormsTwoClass	181	77	23	0	10	0